

Your clothes may soon change colours on touch

How would you like to wear fabric that could change colours based on sound or touch? Well, the technology now exists. A research called Chromosonic, conducted by Judit Eszter Karpati from Hungary, has made this possible. According to reports, Karpati's efforts to explore how digital media can be mixed with textile medium has resulted in the building of such a medium, which can be called Electronic Programmable Textile Interface.

Karpati apparently used an Arduino with 12V power supply and 20 custom PCB chips, which control four industrial 24V DC power supplies. These tools heat up the two handmade textiles, which have been embedded using nichrome wires and screen-printed thermo-chromatic dyeing.

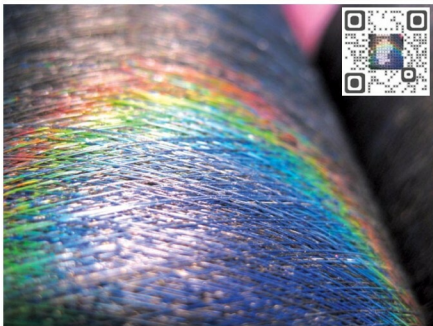
According to reports, the fabric changes its colour based on sound and touch. Many have already said that wearing this could make the wearer look like a chameleon, although the technology would perhaps need a lot more work before that kind of perfection is achieved.

Please scan the QR code (see pic) that describes the project and will help you understand more about it.

Your body heat to charge your mobile phone

In a bid to ensure a stable and reliable power supply to your ever-power-hungry smartphone, a team of researchers has now come up with the idea of a glass-fabric-based thermoelectric (TE) generator. The TE generator will be an extremely light and flexible solution to powering up heart monitors, smart glasses and other wearable tech, producing electricity from the heat emitted from a human body.

"Mobile phones consume high electrical energy compared to electri-



Temperature-sensitive fibre changing colour

cal sensors. Right now we are trying to make a sample that provides electricity for medical sensors," Byung Jin Cho, a professor of electrical engineering at Korea Advanced Institute of Science and Technology in South Korea was quoted as saying. "After that, smartphones will be the next application of the TE generator," he added.

The flexible TE generator could minimise thermal energy loss while maximising power output. It has a self-sustaining structure, eliminating thick external substrates (usually made of ceramic or alumina) that hold inorganic TE materials, and efficiently uses the small yet significant temperature difference between skin and air to create power.

The researchers have developed two types of TE generators. The organic-based TE generators make use of polymers that are highly flexible and compatible with human skin. Although ideal for wearable electronics, these have a low power output. Inorganic-based TE generators, however,

produce a high electrical energy. The downside is they are heavy, rigid and bulky. Cho's technology successfully combines the best of both organic and inorganic tech.

Method to commercialise graphene for electronics revealed

Samsung has discovered a synthesis method to speed up commercialisation of graphene, a unique material ideally suited for electronic devices. Samsung Advanced Institute of Technology (SAIT) has joined hands with Sungkyunkwan University to be the first in the world to develop this new method.

Graphene has one hundred times greater electron mobility than silicon, which is the most widely used material in semiconductors today. It is more durable than steel and has high heat conductivity as well as flexibility, which makes it the perfect material for use in flexible displays, wearables and other next-generation electronic devices.

SAIT revealed a new method of